

Inference for Categorical Time Series with Missing Observations

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Outline

Categorical Time Series

Estimation — Complete Data

Missing Data

Serial Dependence under Missingness

Simulation Study

Summary

Motivation

- ▶ Many real-world time series are **categorical**:
 - ▶ Credit ratings (AAA, AA, ...)
 - ▶ Weather states (sunny, rainy, ...)
 - ▶ Survey responses (agree, neutral, disagree)
- ▶ Classical time series methods assume **numeric structure**
- ▶ $\Rightarrow$ Need new tools for:
 - ▶ Computing proper tests for marginal features
 - ▶ Measuring temporal dependence without bias

Categorical Time Series

Definition

A categorical time series (X_t) takes values in a finite set

$$S = \{s_0, \dots, s_m\}.$$

Two types of categorical data:

- ▶ **Nominal:** no natural ordering
- ▶ **Ordinal:** inherent ordering

Key implication:

- ▶ Available statistical tools depend on structure of S

Marginal Distributions

Central object: marginal probabilities

Nominal data:

$$p_i = P(X_t = s_i)$$

- ▶ Probability Mass Function (PMF)

Ordinal data:

$$f_i = P(X_t \leq s_i)$$

- ▶ Cumulative Distribution Function (CDF)

Key difference:

- ▶ Ordinal $\Rightarrow$ uses ordering
- ▶ Nominal $\Rightarrow$ no cumulative structure

Location Measures

Nominal data:

- ▶ No ordering $\Rightarrow$ no mean/median
- ▶ Use **mode**:

$$\text{Mod}(X_t) = \arg \max_{s_i} P(X_t = s_i)$$

Ordinal data:

- ▶ Ordering allows additional measures
- ▶ **Median**:

$$\text{Med}(X_t) = \min\{s_i : F(s_i) \geq 1/2\}$$

Interpretation:

- ▶ Mode = most frequent category
- ▶ Median = central position in ordering

Dispersion Measures

Different notions of variability

Both indices are scaled to $[0, 1]$, with 0 = minimal dispersion (mass on single category) and 1 = maximal dispersion (differs between types).

Ordinal: Index of Ordinal Variation (IOV)

$$\text{IOV}(\mathbf{f}) = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i)$$

Nominal: Index of Qualitative Variation (IQV)

$$\text{IQV}(\mathbf{p}) = \frac{m+1}{m} \left(1 - \sum_{i=0}^m p_i^2 \right)$$

Stationarity

Why do we need stationarity?

- ▶ Stable probabilistic structure over time
- ▶ Enables estimation and asymptotics

Strict Stationarity

$$F_{X_t, \dots, X_{t+j}} = F_{X_{t+h}, \dots, X_{t+h+j}}$$

for all t, j, h .

- ▶ Distribution invariant under time shifts

Temporal Dependence

Goal: capture dependence across time

Problem:

- ▶ No meaningful arithmetic operations
- ▶ Classical ACF not applicable

Solution:

- ▶ Work with joint probabilities

Bivariate Probabilities

$$p_{ij}(h) = P(X_t = s_i, X_{t+h} = s_j), \quad f_{ij}(h) = P(X_t \leq s_i, X_{t+h} \leq s_j)$$

- ▶ Independence $\Rightarrow p_{ij}(h) = p_i p_j$ and $f_{ij}(h) = f_i f_j$

Cohen's κ

Idea: measure deviation from independence

$$\kappa(h)_{\text{nom}} = \frac{\sum_{j=0}^m (p_{jj}(h) - p_j^2)}{1 - \sum_{i=0}^m p_i^2} \quad \kappa(h)_{\text{ord}} = \frac{\sum_{j=0}^m (f_{jj}(h) - f_j^2)}{1 - \sum_{i=0}^m f_i(1 - f_i)}$$

- ▶ $\kappa = 1$: perfect persistence
- ▶ $\kappa = 0$: independence
- ▶ $\kappa < 0$: tendency to switch states

Interpretation:

- ▶ Probability of staying in same category
- ▶ Relative to independence benchmark

From Theory to Estimation

All quantities defined so far are population quantities.

Central question

How do we estimate $\mathbf{f}$, IOV, and $\kappa_{\text{ord}}(h)$ from data — and what are the statistical properties of these estimators?

Two steps:

1. Estimate marginal CDF $\mathbf{f}$ via relative frequencies
2. Derive distributions of plug-in estimators via the **delta method**

⇒ Key tool: a CLT for dependent data under **mixing conditions**

Estimation of Marginal Probabilities

Idea: use of indicator functions

$$Y_{t,i} = \mathbb{I}\{X_t = s_i\} \quad \text{and} \quad Z_{t,i} = \mathbb{I}\{X_t \leq s_i\}$$

Then estimators are given by:

- ▶ $\hat{p}_i = \frac{1}{n} \sum_{t=1}^n Y_{t,i}$
- ▶ $\hat{f}_i = \frac{1}{n} \sum_{t=1}^n Z_{t,i}$

Intuition:

⇒ we just count relative frequencies

From here we mainly look at ordinal data, but the nominal case is in many ways analogous.

Mixing Assumptions

Idea

The Central Limit Theorem (CLT) provides asymptotic normality and enables statistical inference.

Problem

Classical CLT results assume independence, but in practice data may exhibit serial dependence.

Solution

To handle dependence, we impose **mixing conditions**:

- ▶ α -mixing quantifies the decay of dependence between past and future observations
- ▶ Exponential decay of the mixing coefficients allows a CLT for dependent data

CLT under Serial Dependence

Result

$$\sqrt{n}(\hat{f} - f) \Rightarrow \mathcal{N}(0, \Sigma)$$

where the elements of Σ are given by:

$$\sigma_{ij} = \underbrace{(f_{\min\{i,j\}} - f_i f_j)}_{\text{lag}(0)} + \sum_{h=1}^{\infty} \underbrace{(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)}_{\text{lag}(h)}.$$

Note that $\text{lag}(h) \rightarrow 0$ as $h \rightarrow \infty$.

CLTs for Marginal Estimators

Idea

IOV is a function of the marginal probabilities

⇒ apply the **delta method** to obtain asymptotic distributions

Results

- Asymptotic variance:

$$\sqrt{n} \text{Var}(\widehat{\text{IOV}}) \stackrel{\text{asy}}{=} \frac{16}{m^2} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) \sigma_{ij}$$

- Bias to order n^{-1} :

$$n \text{Bias}(\widehat{\text{IOV}}) = -\frac{4}{m} \sum_{i=0}^{m-1} \sigma_{ii}$$

Why Does Missingness Matter?

Missing observations are ubiquitous in practice:

- ▶ Panel surveys: participants drop out over time
- ▶ Administrative data: recording gaps or system failures
- ▶ Sensor data: transmission errors, device downtime

Key problem: naive estimators are biased

- ▶ Simply ignoring missing observations distorts relative frequencies
- ▶ $\Rightarrow$ biased estimates of $\mathbf{f}$, IOV, and $\kappa_{\text{ord}}(h)$

Goal:

- ▶ Correct for missingness — under suitable assumptions
- ▶ Derive valid asymptotic theory for the corrected estimators

Amplitude Modulation

Idea

Missingness can be modeled as a binary process (O_t) , that *modulates* the *amplitude* of the actual process:

$$O_t = 1 \text{ if } X_t \text{ is observed and } O_t = 0 \text{ if } X_t \text{ is missing}$$

The observed process is then given by $(O_t X_t)$.

Assumptions

- ▶ (O_t) is stationary with $P(O_t = 1) = \pi > 0$ and joint moments $E[O_t O_{t+h}] = \pi(h)$
- ▶ (O_t) is α -mixing with exponential decay
- ▶ (O_t) and (X_t) are **independent**

Missingness Mechanisms

Originally developed for cross-sectional data, but also relevant for time series:

- ▶ **MCAR:** Missing Completely At Random

$$P(O_t = 1 | X_t, X_{t-1}, \dots) = P(O_t = 1)$$

- ▶ **MAR:** Missing At Random

$$P(O_t = 1 | X_t, X_{t-1}, \dots) = P(O_t = 1 | X_{t-1}, \dots)$$

- ▶ **MNAR:** Missing Not at Random

$$P(O_t = 1 | X_t, X_{t-1}, \dots) \neq P(O_t = 1 | X_{t-1}, \dots)$$

From here on, unless stated otherwise, we assume MCAR.

Naive vs. Corrected Estimators

Naive estimator

$$\hat{\mathbf{f}} = \frac{1}{n} \sum_{t=1}^n O_t \mathbf{Z}_t$$

$\Rightarrow$ biased and inconsistent: $E[\hat{f}_i] = \pi f_i \neq f_i$

Corrected estimator

$$\hat{\mathbf{f}}^* = \frac{\sum_{t=1}^n O_t \mathbf{Z}_t}{\sum_{t=1}^n O_t}$$

$\Rightarrow$ divide by number of observed time points, not by n

Asymptotic Properties of the Corrected Estimator

Result

$$\sqrt{n}(\hat{\mathbf{f}}^* - \mathbf{f}) \Rightarrow \mathcal{N}(0, \mathbf{\Sigma}^*)$$

where the elements of $\mathbf{\Sigma}^*$ are given by:

$$\sigma_{ij}^* = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \frac{1}{\pi^2} \sum_{h=1}^{\infty} \pi(h)(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)$$

Remarks:

- ▶ Inflation factor $1/\pi$ at lag 0: fewer effective observations
- ▶ Serial dependence in (O_t) enters via $\pi(h) = E[O_t O_{t+h}]$
- ▶ Reduces to $\mathbf{\Sigma}$ when $\pi = 1$ (no missing data)

Marginal Statistics under Missingness

Strategy

Replace Σ by Σ^* in all delta-method results

IOV (nonlinear in $\mathbf{f}$, delta method applies):

$$\sqrt{n} \text{Var}(\text{IOV}(\hat{\mathbf{f}}^*)) \stackrel{\text{asy.}}{=} \frac{16}{m^2} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) \sigma_{i,j}^*$$

$$\text{Bias}(\text{IOV}(\hat{\mathbf{f}}^*)) = -\frac{4}{mn} \sum_{i=0}^{m-1} \sigma_{i,i}^*$$

Skew (linear in $\mathbf{f}$, Hessian vanishes):

- ▶ Estimator is **unbiased**
- ▶ Asymptotic variance analogous, replacing Σ by Σ^*

Estimating Bivariate Probabilities

Need to estimate $f_{ij}(h) = P(X_t \leq s_i, X_{t+h} \leq s_j)$

Naive estimator:

$$\hat{f}_{ij}(h) = \frac{1}{n-h} \sum_{t=h+1}^n O_t O_{t-h} Z_{t,i} Z_{t-h,j}$$

Problem: $E[\hat{f}_{ij}(h)] = \pi(h) \cdot f_{ij}(h) \neq f_{ij}(h)$

Corrected estimator:

$$\hat{f}_{ij}^*(h) = \frac{\frac{1}{n-h} \sum_{t=h+1}^n O_t O_{t-h} Z_{t,i} Z_{t-h,j}}{\frac{1}{n-h} \sum_{t=h+1}^n O_t O_{t-h}} =: \hat{f}_{ij}(h) / \overline{OO}_h$$

$\Rightarrow$ divide by the proportion of **jointly observed** pairs at lag h

Estimating Cohen's κ under Missingness

Plug-in estimator:

$$\hat{\kappa}_{\text{ord}}(h) = \frac{\sum_{j=0}^{m-1} \left(\hat{f}_{jj}^*(h) - (\hat{f}_j^*)^2 \right)}{\sum_{i=0}^{m-1} \hat{f}_i^* (1 - \hat{f}_i^*)}$$

Joint asymptotic distribution via delta method:

- ▶ Define augmented vector

$$\hat{\mathbf{f}}^{(h)*} = (\hat{f}_0^*, \dots, \hat{f}_{m-1}^*, \hat{f}_{00}^*(h), \dots, \hat{f}_{m-1, m-1}^*(h))^{\top}$$

- ▶ CLT holds for the sample mean of the underlying binary process $\mathbf{Z}_t^{(h)*}$
- ▶ Apply delta method to κ_{ord}

Key challenge: long-run covariance involves third- and fourth-order mixed moments of (X_t) — not closed-form in general.
 $\Rightarrow$ focus on the **null distribution** first

Asymptotic Null Distribution of $\hat{\kappa}_{\text{ord}}(h)$

Under H_0 : (X_t) is i.i.d., $f_{ij}(h) = f_i f_j$, and higher-order moments factor.

Theorem

Under H_0 and MCAR,

$$\sqrt{n} \hat{\kappa}_{\text{ord}}(h) \xrightarrow{d} \mathcal{N}(0, \sigma_{\kappa}^2),$$

where

$$\sigma_{\kappa}^2 = \frac{\frac{1}{\pi(h)} \sum_{i,j=0}^{m-1} (f_{\min\{i,j\}} - f_i f_j) \left(f_{\min\{i,j\}} + \left(1 + \frac{2\pi(h, 2h)}{\pi(h)} - \frac{4\pi(h)}{\pi} \right) f_i f_j \right)}{\left(\sum_{k=0}^{m-1} f_k (1 - f_k) \right)^2}$$

with $\pi(h, 2h) = E[O_t O_{t+h} O_{t-h}]$.

Bias Correction for $\hat{\kappa}_{\text{ord}}(h)$

Second-order Taylor expansion yields the finite-sample bias under H_0 :

$$\text{Bias}(n\hat{\kappa}_{\text{ord}}(h)) \approx -\frac{1}{\pi}$$

Bias-corrected estimator:

$$\hat{\kappa}_{\text{ord}}^{\text{bc}}(h) := \hat{\kappa}_{\text{ord}}(h) + \frac{1}{n\pi}$$

Remarks:

- ▶ Bias increases as $\pi \downarrow$ (more missingness)
- ▶ Under complete data ($\pi = 1$): reduces to the known $-1/n$ bias
- ▶ Closed-form correction — no resampling needed

Consistency under H_A

Under H_A : (X_t) is not i.i.d., so $\kappa_{\text{ord}}(h) \neq 0$ for some h .

Result

$\hat{\kappa}_{\text{ord}}(h)$ is a continuous function of $\hat{f}_i^*$ and $\hat{f}_{ii}^*(h)$.

By the **continuous mapping theorem**:

$$\hat{\kappa}_{\text{ord}}(h) \xrightarrow{P} \kappa_{\text{ord}}(h) \neq 0,$$

and therefore

$$\sqrt{n} \hat{\kappa}_{\text{ord}}(h) \xrightarrow{P} \infty.$$

$\Rightarrow$ The test is **consistent**: rejection probability $\rightarrow 1$ under H_A .

Simulation Goals and Setup

Three goals:

1. Assess finite-sample accuracy of asymptotic bias formulas
2. Examine convergence to asymptotic normality
3. Study robustness to violations of the MCAR assumption

Data generating process: BinAR(1) model

- ▶ Flexible ordinal time series with controllable serial dependence
- ▶ Stationary marginal distribution: $X_t \sim \text{Binomial}(m, p)$
- ▶ Parameter $r \in (-1, 1)$ controls autocorrelation ($r = 0$: i.i.d.)
- ▶ All quantities (IOV, κ_{ord}) available in **closed form**

Simulation Scenarios

Two scenarios chosen to differ qualitatively:

Scenario A

$m = 3, p = 0.20, r = 0.35$

- ▶ Few categories
- ▶ Skewed marginal distribution
- ▶ Moderate serial dependence

Scenario B

$m = 10, p = 0.45, r = 0.50$

- ▶ Many categories
- ▶ Near-symmetric distribution
- ▶ Strong serial dependence

Design:

- ▶ Observation rates: $\pi \in \{1, 0.75\}$
- ▶ Sample sizes: $n \in \{50, 100, 250, 500, 1000\}$
- ▶ $B = 1,000$ Monte Carlo replications

Missingness Mechanisms in the Simulation

Four observation processes (O_t) are studied:

1. **Independent MCAR:** $O_t \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\pi)$
Baseline; satisfies all theoretical assumptions.
2. **Dependent MCAR:** (O_t) follows BinAR(1) with $m = 1$
Serial dependence in missingness; MCAR still holds marginally.
3. **MNAR — increasing:** $\pi_t = \pi_{\text{low}} + (\pi_{\text{high}} - \pi_{\text{low}}) \frac{X_t}{m}$
Higher categories more likely to be observed.
4. **MNAR — decreasing:** $\pi_t = \pi_{\text{high}} - (\pi_{\text{high}} - \pi_{\text{low}}) \frac{X_t}{m}$
Lower categories more likely to be observed.

Performance Measures

For each estimator $\hat{\theta}$, we compute across B replications:

- ▶ **Mean bias:** $\frac{1}{B} \sum_{b=1}^B (\hat{\theta}_b - \theta)$
Compared against theoretical $O(n^{-1})$ formula.
- ▶ **n -scaled bias:** $n \cdot \frac{1}{B} \sum_{b=1}^B (\hat{\theta}_b - \theta)$
Should converge to the theoretical bias limit as $n \rightarrow \infty$.
- ▶ **Standard deviation:** compared against asymptotic SD
 $\sqrt{\sigma_{\hat{\theta}}^2/n}$

Theoretical benchmarks:

- ▶ IOV bias limit: $-\frac{4}{mn} \text{tr}(\mathbf{\Sigma}^*)$
- ▶ κ bias limit (under H_0): $-\frac{1}{n\pi}$

Summary

- ▶ Categorical time series require new tools:
 - ▶ Nominal vs. ordinal $\Rightarrow$ PMF vs. CDF, mode vs. median
- ▶ Marginal estimation:
 - ▶ Relative frequency estimators + CLT under α -mixing
 - ▶ Delta method for IOV and skew
- ▶ Missingness:
 - ▶ Amplitude modulation framework under MCAR
 - ▶ Corrected estimators $\hat{f}^*$ and $\hat{f}_{ij}^*(h)$
 - ▶ Full asymptotic theory carries over via delta method
- ▶ Serial dependence under missingness:
 - ▶ Asymptotic null distribution of $\hat{\kappa}_{\text{ord}}(h)$
 - ▶ Closed-form bias correction $\hat{\kappa}_{\text{ord}}^{\text{bc}}(h) = \hat{\kappa}_{\text{ord}}(h) + \frac{1}{n\pi}$
 - ▶ Consistency under H_A
- ▶ Simulation study validates finite-sample behavior and robustness